A guide to the arithmetic certificate for $K = \mathbf{Q}(\sqrt{-35663739})$, $p = 5$

Generated by `tools/certificate_guide.gp`

What this document is

This guide restates the content of `certificate.gp` in the notation of Section 2.5 of the paper. Each of the 18 entries certifies one value $D_x(e_j)$ of a secondary norm operator: it stores the unramified cyclic quintic L_x/K, the normalized generator σ_x, the pair (a', J) representing the torsion class e_j, the auxiliary ideal I', the compactly represented element t, a residue prime for the sign check, and the expected norm class $[N_x(I')]$. The two defining equations are

$$I'^{\,(1-\sigma_x)^2}\,(t)\,J\mathcal{O}_{L_x} = \mathcal{O}_{L_x}, \qquad\qquad N_x(t) = a'.$$

Facts marked **Checked** are recomputed exactly by the generating script; the full verification is performed by the shared verifier `certificate/verify_certificate.c` as described in the README files of the `certificate` directory.

Notation and conventions

Ideals as matrices. An ideal of the ring of integers is stored by its *Hermite normal form* (HNF): an upper-triangular integer matrix whose columns are the coordinates, with respect to the fixed integral basis, of a $\mathbf{Z}$-basis of the ideal. For example, the ideal $J = \left(\begin{smallmatrix} 3255 & 1924 \\ 0 & 1 \end{smallmatrix}\right)$ of Entry 1 below, over K with integral basis $(1, s)$, denotes the ideal $\mathbf{Z}\,3255 + \mathbf{Z}\,(1924 + s)$. The determinant of the matrix is the norm of the ideal. Elements are likewise given by their coordinate vectors in the integral basis.

Which integral basis. The one PARI returns for the field in question – and that basis is LLL-reduced, so it is not canonical: another machine may reduce differently, and the same coordinate vector would then denote a different algebraic number. The certificate therefore records the basis it was written against, as its last field per entry and in the header for K. The verifier expresses the stored basis in its own, checks that the resulting matrix is integral with determinant ± 1 – so that both bases span the same ring of integers – and converts the coordinates before checking anything. Where the two bases agree, that matrix is the identity and nothing changes.

Characters. The header fixes three generators $\kappa_1, \kappa_2, \kappa_3$ of $\mathrm{Cl}(K)$ (as HNF ideals). A character x on $\mathrm{Cl}(K)/5$ is recorded by its value vector $(x(\bar\kappa_1), x(\bar\kappa_2), x(\bar\kappa_3))$; thus $(1, 0, 0)$ is the character dual to $\bar\kappa_1$. The column number j records which basis element e_j of $\mathrm{Cl}(K)[5]$ the stored pair (a', J) represents.

Automorphisms. σ_x is a field automorphism of L_x. Writing θ for the chosen root of the stored absolute polynomial, so that $L_x = \mathbf{Q}(\theta)$, the automorphism is determined by the single value $\sigma_x(\theta)$, and the certificate stores the coordinate vector of $\sigma_x(\theta)$ in the integral basis of L_x.

Base field data

The base field is $K = \mathbf{Q}(s)$ with $s^2 - s + 8915935 = 0$, of discriminant -35663739. Class group invariants $(30 \quad 10 \quad 5)$, class number 1500; the three stored generator ideals (HNF) are $\begin{pmatrix} 137 & 94 \\ 0 & 1 \end{pmatrix}$, $\begin{pmatrix} 1605 & 490 \\ 0 & 1 \end{pmatrix}$, $\begin{pmatrix} 655 & 269 \\ 0 & 1 \end{pmatrix}$, and the torsion units are generated by -1 of order 2.

Entry 1: character a, column e_1

Prescribed character vector $(1 \quad 0 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 + 332y^3 + (2s - 1)\, y^2 + 173502y + (1224s - 612)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 + 664*y^8 + 457228*y^6 + 150869067*y^4 + 73755360540*y^2 + 1335
7639460016
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad 0 \quad -1 \quad -1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{440032}{1159963732423125} + (\frac{21893}{405987306348093754})\, s, \qquad J = \begin{pmatrix} 3255 & 1924 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 3255$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $10565306426304 = 2^6 \cdot 3^2 \cdot 7^2 \cdot 17 \cdot 313 \cdot 70351$:

$$\begin{pmatrix}
31444364364 & 20202387876 & 12851504618 & 8259209223 & 15120591628 & 6779836833 & 9141892896 & 14037702504 & 18289279276 & 29881837590 \\
0 & 12 & 2 & 1 & 4 & 8 & 5 & 11 & 5 & 3 \\
0 & 0 & 14 & 3 & 2 & 11 & 1 & 10 & 1 & 13 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 2 & 0 & 1 & 1 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 197 factors with signed exponents of absolute value up to 1576961670; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (0 \quad 0 \quad 4)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 2: character a, column e_2

Prescribed character vector $(1 \quad 0 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 + 332y^3 + (2s - 1)\, y^2 + 173502y + (1224s - 612)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 + 664*y^8 + 457228*y^6 + 150869067*y^4 + 73755360540*y^2 + 1335
7639460016
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad 0 \quad -1 \quad -1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{1879447}{104074241381523} + (-\frac{949}{104074241381523})\, s, \qquad J = \begin{pmatrix} 987 & 238 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 987$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $20802457174464 = 2^6 \cdot 3^5 \cdot 7^2 \cdot 197 \cdot 138569$:

$$\begin{pmatrix}
2293039812 & 1523791752 & 26025892 & 1247666655 & 1397914383 & 67341697 & 2024433285 & 1971708712 & 1755460136 & 2265108207 \\
0 & 42 & 18 & 27 & 27 & 24 & 21 & 20 & 23 & 32 \\
0 & 0 & 4 & 3 & 3 & 3 & 0 & 0 & 3 & 0 \\
0 & 0 & 0 & 3 & 0 & 0 & 0 & 0 & 0 & 1 \\
0 & 0 & 0 & 0 & 3 & 0 & 0 & 2 & 2 & 1 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 3 & 0 & 0 & 2 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 1 & 1 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 191 factors with signed exponents of absolute value up to 57315; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (0 \quad 3 \quad 2)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 3: character a, column e_3

Prescribed character vector $(1 \quad 0 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 + 332y^3 + (2s - 1)\, y^2 + 173502y + (1224s - 612)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 + 664*y^8 + 457228*y^6 + 150869067*y^4 + 73755360540*y^2 + 1335
7639460016
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad 0 \quad -1 \quad -1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{1695709}{24112181031875} + (-\frac{2337}{120560905159375})\, s, \qquad J = \begin{pmatrix} 655 & 269 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 655$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $2917152748908 = 2^2 \cdot 3^4 \cdot 17 \cdot 47 \cdot 1171 \cdot 9623$:

$$\begin{pmatrix}
162064041606 & 137019750354 & 8473130983 & 52420887168 & 67359498522 & 79448464693 & 122571838641 & 18593897230 & 97785198228 & 8468098328 \\
0 & 6 & 3 & 3 & 3 & 3 & 2 & 3 & 4 & 1 \\
0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 3 & 1 & 0 & 1 & 0 & 2 & 1 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 188 factors with signed exponents of absolute value up to 1060; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (0 \quad 0 \quad 0)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 4: character b, column e_1

Prescribed character vector $(0 \quad 1 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 + 1553y^3 + (8s - 4)\, y^2 + 2340789y + (-9576s + 4788)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 + 3106*y^8 + 7093387*y^6 + 7841110458*y^4 + 4113229283865*y^2 +
817589219405616
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(2 \quad -1 \quad 2 \quad -1 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{440032}{1159963732423125} + \left(\frac{21893}{40598730634809375}\right) s, \qquad J = \begin{pmatrix} 3255 & 1924 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 3255$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $25036491035817 = 3^3 \cdot 107 \cdot 409 \cdot 21188617$:

$$\begin{pmatrix}
2781832337313 & 2077812088608 & 1360140153417 & 2366298449845 & 331118006177 & 1129823163590 & 59528349723 & 562953834221 & 1456307020278 & 2005627194803 \\
0 & 3 & 0 & 1 & 2 & 0 & 0 & 0 & 2 & 2 \\
0 & 0 & 3 & 2 & 1 & 2 & 0 & 1 & 2 & 0 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 250 factors with signed exponents of absolute value up to 1197086971996137945616 05; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (1 \quad 0 \quad 1)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 5: character b, column e_2

Prescribed character vector $(0 \quad 1 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 + 1553y^3 + (8s - 4)\, y^2 + 2340789y + (-9576s + 4788)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 + 3106*y^8 + 7093387*y^6 + 7841110458*y^4 + 4113229283865*y^2 +
817589219405616
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(2 \quad -1 \quad 2 \quad -1 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{1879447}{104074241381523} + (-\frac{949}{104074241381523})\, s, \qquad J = \begin{pmatrix} 987 & 238 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 987$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $2665698605244 = 2^2 \cdot 3^6 \cdot 10789 \cdot 84731$:

$$\begin{pmatrix} 16454929662 & 9971860647 & 8587602123 & 6984204257 & 9048264159 & 13112733836 & 3530574461 & 7780144140 & 7643174006 & 9960270623 \\ 0 & 3 & 0 & 2 & 0 & 0 & 2 & 0 & 0 & 0 \\ 0 & 0 & 3 & 1 & 0 & 2 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 3 & 0 & 1 & 0 & 2 & 2 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 3 & 2 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 245 factors with signed exponents of absolute value up to 519237767452699074055 0; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (2 \quad 0 \quad 2)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 6: character b, column e_3

Prescribed character vector $(0 \quad 1 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 + 1553y^3 + (8s - 4)\, y^2 + 2340789y + (-9576s + 4788)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 + 3106*y^8 + 7093387*y^6 + 7841110458*y^4 + 4113229283865*y^2 +
817589219405616
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(2 \quad -1 \quad 2 \quad -1 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{1695709}{24112181031875} + (-\frac{2337}{120560905159375})\, s, \qquad J = \begin{pmatrix} 655 & 269 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 655$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $504473186448 = 2^4 \cdot 3^2 \cdot 7^2 \cdot 11^2 \cdot 67 \cdot 8819$:

$$\begin{pmatrix} 545966652 & 240672993 & 13897140 & 239940997 & 373501409 & 370017632 & 540437499 & 532675738 & 455483489 & 102061283 \\ 0 & 33 & 11 & 26 & 11 & 11 & 18 & 18 & 17 & 32 \\ 0 & 0 & 7 & 1 & 1 & 3 & 0 & 2 & 0 & 2 \\ 0 & 0 & 0 & 4 & 0 & 0 & 0 & 0 & 0 & 2 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 244 factors with signed exponents of absolute value up to 43458523625733493475426; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (3 \quad 0 \quad 1)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 7: character c, column e_1

Prescribed character vector $(0 \quad 0 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 + (-s - 79)\, y^3 + (34s - 38164)\, y^2 + (-475s + 951764)\, y + (2442s - 8605014)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 159*y^8 - 76294*y^7 + 10825308*y^6 - 617425776*y^5 + 20080859
479*y^4 - 402757877448*y^3 + 5054022702159*y^2 - 37057551106554*y +
127194218301348
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 1 \quad 0 \quad 0 \quad 0 \quad -1 \quad 0 \quad -1 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{440032}{1159963732423125} + \left(\frac{21893}{40598730634809375}\right) s, \qquad J = \begin{pmatrix} 3255 & 1924 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 3255$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $7736623235829 = 3^6 \cdot 10612651901$:

$$\begin{pmatrix} 95513867109 & 35395789323 & 13201038366 & 46711363500 & 12949911567 & 16406697115 & 32289976453 & 80650029660 & 59050582979 & 81915923263 \\ 0 & 3 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 \\ 0 & 0 & 3 & 0 & 0 & 2 & 1 & 0 & 1 & 2 \\ 0 & 0 & 0 & 3 & 0 & 1 & 0 & 2 & 1 & 0 \\ 0 & 0 & 0 & 0 & 3 & 2 & 0 & 2 & 0 & 2 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 260 factors with signed exponents of absolute value up to 12342809; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (2 \quad 0 \quad 0)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 8: character c, column e_2

Prescribed character vector $(0 \quad 0 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 + (-s - 79)\, y^3 + (34s - 38164)\, y^2 + (-475s + 951764)\, y + (2442s - 8605014)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 159*y^8 - 76294*y^7 + 10825308*y^6 - 617425776*y^5 + 20080859
479*y^4 - 402757877448*y^3 + 5054022702159*y^2 - 37057551106554*y +
127194218301348
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 1 \quad 0 \quad 0 \quad 0 \quad -1 \quad 0 \quad -1 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{1879447}{104074241381523} + (-\frac{949}{104074241381523})\, s, \qquad J = \begin{pmatrix} 987 & 238 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 987$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $73658776572 = 2^2 \cdot 3^4 \cdot 43 \cdot 5287021$:

$$\begin{pmatrix} 1364051418 & 18911010 & 167500668 & 358584765 & 313568239 & 618976896 & 1358953953 & 219145074 & 637273525 & 1031816654 \\ 0 & 6 & 0 & 3 & 4 & 5 & 5 & 3 & 1 & 4 \\ 0 & 0 & 3 & 0 & 1 & 2 & 2 & 2 & 2 & 0 \\ 0 & 0 & 0 & 3 & 0 & 0 & 1 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 255 factors with signed exponents of absolute value up to 1882221859; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (3 \quad 1 \quad 0)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 9: character c, column e_3

Prescribed character vector $(0 \quad 0 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 + (-s - 79)\, y^3 + (34s - 38164)\, y^2 + (-475s + 951764)\, y + (2442s - 8605014)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 159*y^8 - 76294*y^7 + 10825308*y^6 - 617425776*y^5 + 20080859
479*y^4 - 402757877448*y^3 + 5054022702159*y^2 - 37057551106554*y +
127194218301348
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 1 \quad 0 \quad 0 \quad 0 \quad -1 \quad 0 \quad -1 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{1695709}{24112181031875} + (-\frac{2337}{120560905159375})\, s, \qquad J = \begin{pmatrix} 655 & 269 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 655$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $3456311578704 = 2^4 \cdot 3^3 \cdot 1571 \cdot 5092757$:

$$\begin{pmatrix} 96008654964 & 38110457248 & 18668230464 & 21763382877 & 27962968834 & 23439492850 & 12875245230 & 8233796532 & 53052385570 & 3048570973 \\ 0 & 4 & 0 & 3 & 1 & 1 & 2 & 3 & 2 & 1 \\ 0 & 0 & 3 & 0 & 0 & 0 & 1 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 & 1 & 2 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 255 factors with signed exponents of absolute value up to 2974198877; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (1 \quad 4 \quad 0)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 10: character $a + b$, column e_1

Prescribed character vector $(1 \quad 1 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 2y^4 + 718y^3 + (24s + 75601)\, y^2 + (426s - 1259150)\, y + (-252s - 23763152)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 4*y^9 + 1440*y^8 + 148354*y^7 - 2304802*y^6 + 66089460*y^5 +
9140123765*y^4 - 42195040202*y^3 - 498522080904*y^2 + 57918652976408
*y + 565259578825648
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad 0 \quad 0 \quad -1 \quad 0 \quad 0 \quad 0 \quad -1 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{440032}{1159963732423125} + \left(\frac{21893}{40598730634809375}\right) s, \qquad J = \begin{pmatrix} 3255 & 1924 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 3255$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $9025762339584 = 2^8 \cdot 3^2 \cdot 19^2 \cdot 10851611$:

$$\begin{pmatrix}
2474167308 & 1164074292 & 184340060 & 359135148 & 2315003740 & 2301188020 & 1215514084 & 2276873455 & 1165896335 & 647190079 \\
0 & 76 & 64 & 0 & 53 & 60 & 7 & 74 & 43 & 39 \\
0 & 0 & 4 & 0 & 1 & 3 & 0 & 3 & 0 & 1 \\
0 & 0 & 0 & 12 & 0 & 1 & 0 & 1 & 0 & 4 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 177 factors with signed exponents of absolute value up to 187471390; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (1 \quad 4 \quad 0)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 11: character $a + b$, column e_2

Prescribed character vector $(1 \quad 1 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 2y^4 + 718y^3 + (24s + 75601)\, y^2 + (426s - 1259150)\, y + (-252s - 23763152)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 4*y^9 + 1440*y^8 + 148354*y^7 - 2304802*y^6 + 66089460*y^5 +
9140123765*y^4 - 42195040202*y^3 - 498522080904*y^2 + 57918652976408
*y + 565259578825648
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad 0 \quad 0 \quad -1 \quad 0 \quad 0 \quad 0 \quad -1 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{1879447}{104074241381523} + \left(-\frac{949}{104074241381523}\right) s, \qquad J = \begin{pmatrix} 987 & 238 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 987$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $2136229565904 = 2^4 \cdot 3^3 \cdot 7 \cdot 41^2 \cdot 420241$:

$$\begin{pmatrix}
723655002 & 574057044 & 235636758 & 649008388 & 80605386 & 509071573 & 109248393 & 17128073 & 623091001 & 685243890 \\
0 & 6 & 0 & 1 & 0 & 0 & 5 & 1 & 2 & 1 \\
0 & 0 & 246 & 209 & 84 & 231 & 1 & 68 & 73 & 234 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 2 & 0 & 1 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 174 factors with signed exponents of absolute value up to 267645946; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (4 \quad 1 \quad 1)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 12: character $a + b$, column e_3

Prescribed character vector $(1 \quad 1 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 2y^4 + 718y^3 + (24s + 75601)\, y^2 + (426s - 1259150)\, y + (-252s - 23763152)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 4*y^9 + 1440*y^8 + 148354*y^7 - 2304802*y^6 + 66089460*y^5 +
9140123765*y^4 - 42195040202*y^3 - 498522080904*y^2 + 57918652976408
*y + 565259578825648
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad 0 \quad 0 \quad -1 \quad 0 \quad 0 \quad 0 \quad -1 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{1695709}{24112181031875} + (-\frac{2337}{120560905159375})\, s, \qquad J = \begin{pmatrix} 655 & 269 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 655$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $1790353844307 = 3^3 \cdot 7 \cdot 449 \cdot 21097487$:

$$\begin{pmatrix}
198928204923 & 182752009827 & 192410132604 & 101051000233 & 31094290314 & 72156000094 & 21871988757 & 155446740482 & 56780140274 & 115519788615 \\
0 & 3 & 0 & 0 & 0 & 0 & 2 & 1 & 0 & 0 \\
0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 3 & 0 & 1 & 2 & 0 & 1 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 171 factors with signed exponents of absolute value up to 495160; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (3 \quad 2 \quad 4)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 13: character $a + c$, column e_1

Prescribed character vector $(1 \quad 0 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 + 323y^3 + (-8s + 4)\, y^2 + 682920y + (7632s - 3816)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 + 646*y^8 + 1470169*y^6 + 1011786144*y^4 - 622362897792*y^2 + 5
19330231739584
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad -1 \quad 1 \quad -1 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{440032}{1159963732423125} + (\frac{21893}{405987306348093750})\, s, \qquad J = \begin{pmatrix} 3255 & 1924 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 3255$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $1427698351524 = 2^2 \cdot 3 \cdot 17 \cdot 6998521331$:

$$\begin{pmatrix} 713849175762 & 309742215377 & 356881487816 & 310077959300 & 453006801053 & 365069289013 & 689983809939 & 84365081848 & 429588857344 & 336843479018 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 2 factors with signed exponents of absolute value up to 1; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (3 \quad 3 \quad 2)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 14: character $a + c$, column e_2

Prescribed character vector $(1 \quad 0 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 + 323y^3 + (-8s + 4)\, y^2 + 682920y + (7632s - 3816)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 + 646*y^8 + 1470169*y^6 + 1011786144*y^4 - 622362897792*y^2 + 5
19330231739584
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad -1 \quad 1 \quad -1 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{1879447}{104074241381523} + \left(-\frac{949}{104074241381523}\right) s, \qquad J = \begin{pmatrix} 987 & 238 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 987$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $204632784 = 2^4 \cdot 3^3 \cdot 151 \cdot 3137$:

$$\begin{pmatrix}
2842122 & 110097 & 663606 & 1640288 & 2756777 & 2640022 & 26614 & 694045 & 1631768 & 2587573 \\
0 & 3 & 0 & 0 & 2 & 0 & 2 & 1 & 2 & 2 \\
0 & 0 & 6 & 3 & 1 & 4 & 1 & 2 & 0 & 0 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 2 & 1 & 0 & 0 & 1 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 & 1 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 332 factors with signed exponents of absolute value up to 78011678501730645021338488331515; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (1 \quad 2 \quad 4)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 15: character $a + c$, column e_3

Prescribed character vector $(1 \quad 0 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 + 323y^3 + (-8s + 4)\, y^2 + 682920y + (7632s - 3816)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 + 646*y^8 + 1470169*y^6 + 1011786144*y^4 - 622362897792*y^2 + 5
19330231739584
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad -1 \quad 1 \quad -1 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{1695709}{24112181031875} + (-\frac{2337}{120560905159375})\, s, \qquad J = \begin{pmatrix} 655 & 269 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 655$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $587918327652 = 2^2 \cdot 3^3 \cdot 33223 \cdot 163853$:

$$\begin{pmatrix} 32662129314 & 4811323227 & 8541670383 & 28694935295 & 8771928401 & 12623816159 & 14994032186 & 18190866962 & 19896138292 & 13620890396 \\ 0 & 3 & 0 & 1 & 0 & 2 & 1 & 0 & 2 & 0 \\ 0 & 0 & 3 & 2 & 0 & 0 & 0 & 0 & 2 & 2 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 332 factors with signed exponents of absolute value up to 429159415520284177830440707161772; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (1 \quad 4 \quad 4)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 16: character $b + c$, column e_1

Prescribed character vector $(0 \quad 1 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 + 173y^3 + (6s - 3)\, y^2 - 333198y + (-2214s + 1107)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 + 346*y^8 - 636467*y^6 + 205687143*y^4 - 125857647234*y^2 + 437
04093293811
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad 1 \quad 0 \quad -1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{440032}{1159963732423125} + (\frac{21893}{40598730634809375})\, s, \qquad J = \begin{pmatrix} 3255 & 1924 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 3255$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $89913145003200 = 2^6 \cdot 3^5 \cdot 5^2 \cdot 17^2 \cdot 181 \cdot 4421$:

$$\begin{pmatrix} 2448615060 & 2054213640 & 2058072050 & 2432657719 & 1693970510 & 944214447 & 815236219 & 818352107 & 2295049204 & 831016198 \\ 0 & 60 & 4 & 35 & 48 & 29 & 42 & 37 & 39 & 2 \\ 0 & 0 & 34 & 5 & 18 & 19 & 22 & 31 & 13 & 4 \\ 0 & 0 & 0 & 3 & 0 & 1 & 2 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 3 & 0 & 2 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 182 factors with signed exponents of absolute value up to 11959071; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (0 \quad 4 \quad 1)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 17: character $b + c$, column e_2

Prescribed character vector $(0 \quad 1 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 + 173y^3 + (6s - 3)\, y^2 - 333198y + (-2214s + 1107)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 + 346*y^8 - 636467*y^6 + 205687143*y^4 - 125857647234*y^2 + 437
04093293811
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad 1 \quad 0 \quad -1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{1879447}{104074241381523} + (-\frac{949}{104074241381523})\, s, \qquad J = \begin{pmatrix} 987 & 238 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 987$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $89913145003200 = 2^6 \cdot 3^5 \cdot 5^2 \cdot 17^2 \cdot 181 \cdot 4421$:

$$\begin{pmatrix} 2448615060 & 2054213640 & 2058072050 & 2432657719 & 1693970510 & 944214447 & 815236219 & 818352107 & 2295049204 & 831016198 \\ 0 & 60 & 4 & 35 & 48 & 29 & 42 & 37 & 39 & 2 \\ 0 & 0 & 34 & 5 & 18 & 19 & 22 & 31 & 13 & 4 \\ 0 & 0 & 0 & 3 & 0 & 1 & 2 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 3 & 0 & 2 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 189 factors with signed exponents of absolute value up to 11876792; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (0 \quad 4 \quad 1)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 18: character $b + c$, column e_3

Prescribed character vector $(0 \quad 1 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 + 173y^3 + (6s - 3)\,y^2 - 333198y + (-2214s + 1107)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 + 346*y^8 - 636467*y^6 + 205687143*y^4 - 125857647234*y^2 + 437
04093293811
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad 1 \quad 0 \quad -1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{1695709}{24112181031875} + (-\frac{2337}{120560905159375})\,s, \qquad J = \begin{pmatrix} 655 & 269 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 655$. **Checked:** $(a')\,J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $762042390000 = 2^4 \cdot 3 \cdot 5^4 \cdot 911 \cdot 27883$:

$$\begin{pmatrix} 3810211950 & 2324041530 & 3611478250 & 1040602110 & 3697042603 & 2607704572 & 671968116 & 2260715024 & 707808867 & 3473898188 \\ 0 & 10 & 0 & 4 & 4 & 5 & 0 & 6 & 2 & 6 \\ 0 & 0 & 5 & 2 & 2 & 0 & 0 & 1 & 4 & 0 \\ 0 & 0 & 0 & 2 & 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 195 factors with signed exponents of absolute value up to 259033; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (1 \quad 3 \quad 2)$ in the coordinates of $\mathrm{Cl}(K)/5$.